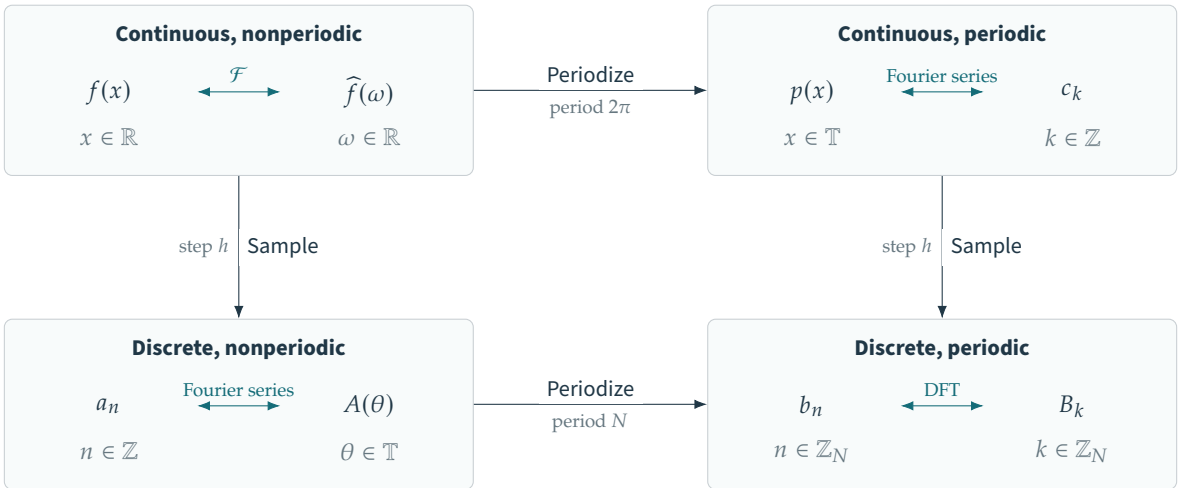
$$\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z}), \qquad \mathbb{Z}_N = \mathbb{Z}/N\mathbb{Z}, \qquad h = 2\pi/N$$



$$p(x) = \sum_{j\in\mathbb{Z}} f(x + 2\pi j), \qquad a_n = f(hn), \qquad b_n = p(hn) = \sum_{j\in\mathbb{Z}} a_{n+jN}$$

Periodization in one domain corresponds to sampling in the other

$$c_k = \frac{\widehat{f}(k)}{2\pi}, \qquad A(\theta) = \sum_{n\in\mathbb{Z}} a_n e^{-in\theta} = \frac{1}{h}\sum_{j\in\mathbb{Z}} \widehat{f}\left(\frac{\theta + 2\pi j}{h}\right)$$

$$B_k = \sum_{n=0}^{N-1} b_n e^{-2\pi i n k/N} = A\left(\frac{2\pi k}{N}\right) = N\sum_{j\in\mathbb{Z}} c_{k+jN}$$

The displayed identities hold, for example, for a rapidly decaying smooth signal $f \in \mathcal{S}(\mathbb{R})$.